

Library Management – OOP Concepts

Java Lab Record

1. Aim

To develop a Java program to demonstrate Object-Oriented Programming concepts including encapsulation using a Book class, inheritance, method overloading, method overriding, abstraction, and interfaces.

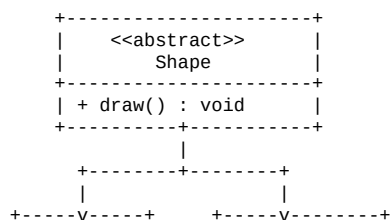
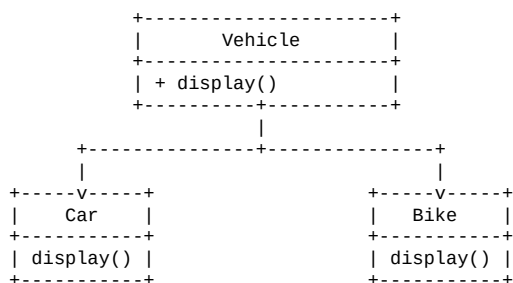
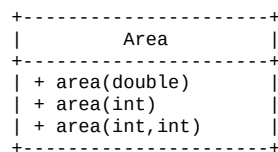
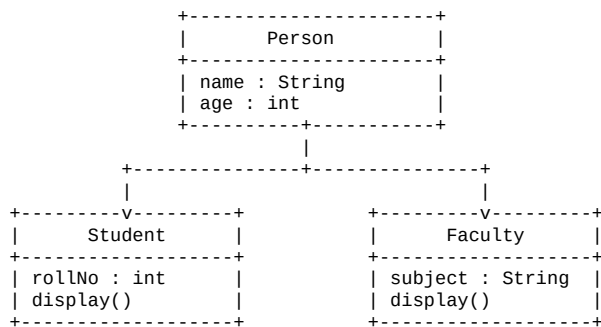
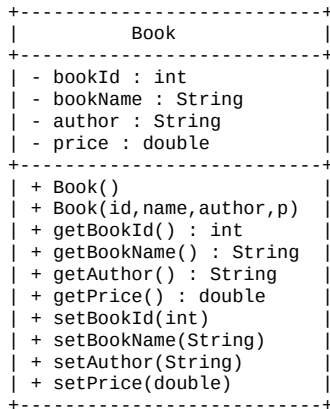
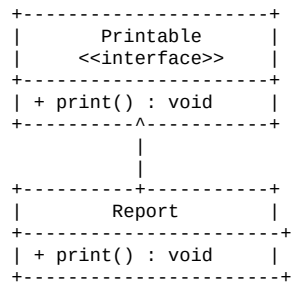
2. Step-by-Step Procedure

1. Create the Printable interface with the print() method.
2. Create the Book class with private fields bookId, bookName, author, and price.
3. Define a default constructor and a parameterized constructor for Book.
4. Create getter and setter methods to access and modify Book data.
5. Create the Person class containing name and age.
6. Create Student and Faculty classes that inherit from Person.
7. Define display() methods in Student and Faculty to display their respective details.
8. Create the Area class with overloaded area() methods for circle, square, and rectangle.
9. Create Vehicle as a base class and override display() in Car and Bike.
10. Create the abstract Shape class with the abstract draw() method.
11. Create Circle and Rectangle classes that implement draw().
12. Create the Report class that implements Printable and defines print().
13. In main(), create objects and invoke the corresponding methods.
14. Display the output of all operations.

3. Algorithm

1. Start.
2. Define the Printable interface.
3. Define the Book class with private data members, constructors, getters, and setters.
4. Define Person and derive Student and Faculty from it.
5. Define the Area class with overloaded area() methods.
6. Define Vehicle and derive Car and Bike with overridden display() methods.
7. Define the abstract Shape class and derive Circle and Rectangle.
8. Define Report and implement Printable.
9. Create a Book object and display its details.
10. Create Student and Faculty objects and display their details.
11. Create an Area object and calculate the area of a circle, square, and rectangle.
12. Create Car and Bike objects and call display().
13. Create Circle and Rectangle objects and call draw().
14. Create a Report object and call print().
15. Display all results.
16. Stop.

4. UML Class Diagram



Circle	Rectangle
+-----+	+-----+
draw()	draw()
+-----+	+-----+

5. Code

```
package core_java;

interface Printable {
    void print();
}

// Part A & B: Book Class
class Book {
    private int bookId;
    private String bookName;
    private String author;
    private double price;

    // Default Constructor
    Book() {
        bookId = 0;
        bookName = "Unknown";
        author = "Unknown";
        price = 0;
    }

    // Parameterized Constructor
    Book(int id, String name, String author, double price) {
        this.bookId = id;
        this.bookName = name;
        this.author = author;
        this.price = price;
    }

    public int getBookId() {
        return bookId;
    }

    public String getBookName() {
        return bookName;
    }

    public String getAuthor() {
        return author;
    }

    public double getPrice() {
        return price;
    }

    public void setBookId(int id) {
        bookId = id;
    }

    public void setBookName(String name) {
        bookName = name;
    }

    public void setAuthor(String a) {
        author = a;
    }

    public void setPrice(double p) {
        price = p;
    }
}

// Part C: Inheritance
class Person {
    String name = "Meghana";
    int age = 19;
}

class Student extends Person {
    int rollNo = 101;

    void display() {
        System.out.println("Student Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Roll No: " + rollNo);
    }
}

class Faculty extends Person {
    String subject = "Java";

    void display() {
        System.out.println("Faculty Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Subject: " + subject);
    }
}

// Part D: Method Overloading
class Area {
    double area(double radius) {
        return 3.14 * radius * radius;
    }

    int area(int side) {
        return side * side;
    }

    int area(int length, int breadth) {
```

```

        return length * breadth;
    }
}

// Method Overriding
class Vehicle {
    void display() {
        System.out.println("This is a Vehicle");
    }
}

class Car extends Vehicle {
    void display() {
        System.out.println("This is a Car");
    }
}

class Bike extends Vehicle {
    void display() {
        System.out.println("This is a Bike");
    }
}

// Part E: Abstraction
abstract class Shape {
    abstract void draw();
}

class Circle extends Shape {
    void draw() {
        System.out.println("Drawing Circle");
    }
}

class Rectangle extends Shape {
    void draw() {
        System.out.println("Drawing Rectangle");
    }
}

// Interface
class Report implements Printable {
    public void print() {
        System.out.println("Printing Report");
    }
}

public class LibraryManagement {
    public static void main(String[] args) {

        // Book Object
        Book b = new Book(101, "Java Programming", "James Gosling", 550);

        System.out.println("Book ID: " + b.getBookId());
        System.out.println("Book Name: " + b.getBookName());
        System.out.println("Author: " + b.getAuthor());
        System.out.println("Price: " + b.getPrice());

        // Inheritance
        Student s = new Student();
        s.display();

        Faculty f = new Faculty();
        f.display();

        // Method Overloading
        Area a = new Area();
        System.out.println("Area of Circle: " + a.area(5.0));
        System.out.println("Area of Square: " + a.area(4));
        System.out.println("Area of Rectangle: " + a.area(5, 6));

        // Method Overriding
        Car K = new Car();
        K.display();

        Bike B = new Bike();
        B.display();

        // Abstraction
        Circle c = new Circle();
        c.draw();

        Rectangle r = new Rectangle();
        r.draw();

        // Interface
        Report rep = new Report();
        rep.print();
    }
}

```

6. Sample Output

The following is the output screenshot supplied with the program.

```
Roll No: 101
Faculty Name: Meghana
Age: 19
Subject: Java
Area of Circle: 78.5
Area of Square: 16
Area of Rectangle: 30
This is a Car
This is a Bike
Drawing Circle
Drawing Rectangle
Printing Report
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> 
```

Output represented by the given code:

```
Book ID: 101
Book Name: Java Programming
Author: James Gosling
Price: 550.0
Student Name: Meghana
Age: 19
Roll No: 101
Faculty Name: Meghana
Age: 19
Subject: Java
Area of Circle: 78.5
Area of Square: 16
Area of Rectangle: 30
This is a Car
This is a Bike
Drawing Circle
Drawing Rectangle
Printing Report
```